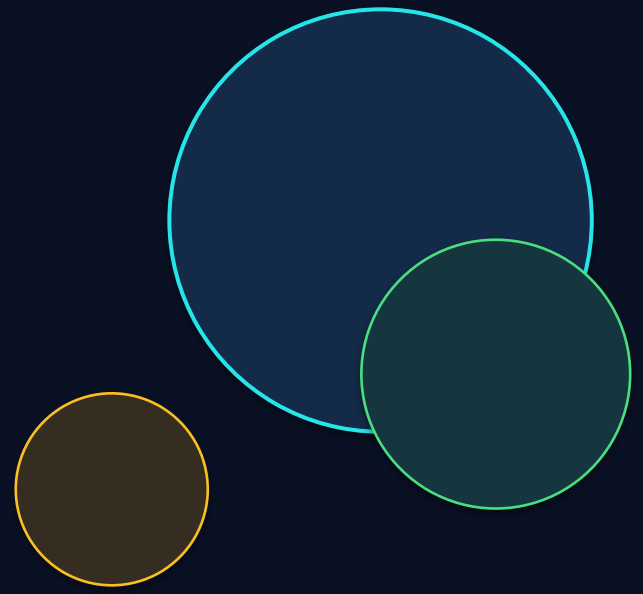
SWIM WITH TINYFISH • HACKATHON PROPOSAL

QuantIntel

Autonomous Agentic Lab for Quantitative Alpha Discovery

Transforming unstructured, live-web corporate disclosures into source-grounded, statistically validated quantitative hypotheses through TinyFish web agents and deterministic Python backtesting.



🌐 LIVE WEB DISCOVERY
TinyFish Agent & Search

📐 PYDANTIC EVIDENCE
Zero-Hallucination Schema

⚡ PYTHON SANDBOX
Deterministic CAR Backtesting

Proposed by: Manish Yadav | Scaler School of Technology • Track: AI Web Agents & FinTech Innovation

The Quantitative Research Chasm

Quantitative researchers spend 70%+ of their time gathering and verifying unstructured web disclosures.

01 • FRAGMENTED LIVE WEB

Alpha Hidden in Dynamic Silos

- **Modern Web is Dynamic:** Market catalysts live behind JavaScript-heavy IR portals, SEC EDGAR tables, FDA registries, and paywalled press releases.
- **Scrapers Break Easily:** Standard HTTP requests fail on single-page apps (SPAs) and trigger anti-bot blocks, starving quant models of timely data.
- **Knowledge Cutoff Limits:** Static LLMs have outdated weights and zero access to live breaking market developments.

02 • THE HALLUCINATION TRAP

Catastrophic Look-Ahead Bias

- **Misattributed Timestamps:** Generic LLMs frequently hallucinate event dates, confusing press release dates with filing dates.
- **The Look-Ahead Trap:** In quantitative backtesting, using information 1 day before it was public invalidates the entire alpha model.
- **Lack of Auditability:** LLM chat outputs provide no click-through source evidence or verifiable provenance for regulatory audit.

03 • THE WEB-TO-CODE GAP

Disjointed Manual Pipelines

- **Text vs Math Disconnect:** Reading news is qualitative and textual; statistical backtesting is quantitative and mathematical.
- **Manual Translation:** Turning a corporate announcement into a rigorous null hypothesis (H0 vs H1) takes hours of manual data collation.
- **Slow Time-to-Validation:** By the time an analyst manually validates a news-driven anomaly, market participants have already arbitraged the alpha.

THE QUANTINTEL OPPORTUNITY: Replace hours of brittle manual web scraping with an autonomous TinyFish agent that extracts traceable, date-verified evidence and outputs executable Python code to backtest alpha hypotheses in minutes.

QuantIntel: Grounded Web-to-Quant Intelligence

An autonomous research lab that pairs live web exploration with deterministic statistical validation.

PILLAR 1

Autonomous Live Web Recon

Navigates the open web using TinyFish

✓ **Live Search & Discovery:** Proactively crawls breaking corporate announcements, earnings calls, and regulatory filings.

✓ **Dynamic Content Extraction:** Renders JavaScript SPAs and converts complex investor portals into clean, low-token Markdown.

✓ **Bypasses Scraping Fragility:** Serverless Chromium browser handles dynamic elements, pagination, and multi-step interactions.

PILLAR 2

Pydantic Evidence Brief

Structures qualitative data with strict auditability

✓ **Schema-Enforced Extraction:** Captures exact UTC event timestamps, validated entity tickers, and full source URLs.

✓ **Citation Audit Trail:** Every single claim links directly to the underlying raw webpage context—zero unsourced statements.

✓ **Falsifiable Hypotheses:** Formulates rigorous statistical tests (e.g., Abnormal Volume Surges, Post-Earnings Volatility) instead of speculation.

PILLAR 3

Deterministic Python Engine

Executes backtests in an isolated sandbox

✓ **Zero-Hallucination Math:** Mathematical calculations and statistical tests are dispatched exclusively to verified Python code.

✓ **Historical Event Studies:** Analyzes Cumulative Abnormal Returns (CAR) and volume z-scores against historical benchmarks (yfinance).

✓ **Academic Rigor:** Reports empirical p-values, t-statistics, and limitations—preventing false discovery and data mining.

OUR CORE PRINCIPLE: The AI Agent proposes structured hypotheses; Python computes the math. QuantIntel does NOT promise guaranteed profits, black-box price predictions, or autonomous trading execution. It delivers empirical, reproducible research.

A Concrete Alpha Research Journey

From open-web catalyst to verifiable statistical output in 4 automated stages.

STAGE 1

The Analyst Prompt

Natural Language Trigger

User Query:
“Investigate recent FDA advisory panel decisions for BioTech tickers ($BIIB, $MRNA). Extract event dates and test if pre-event volume exhibits statistical anomalies.”

Intent Decomposed:

Target: Healthcare/Biotech | Event: FDA AdCom votes | Objective: Volume Event-Study.

STAGE 2

TinyFish Reconnaissance

Autonomous Live Browsing

TinyFish Search:
Discovers official FDA.gov press announcements and BioPharma news wires in real time.

TinyFish Fetch & Agent:

Renders JavaScript calendar tables, navigates PDF docket attachments, and extracts clean markdown text without banner noise.

STAGE 3

Pydantic Structuring

Hypothesis & Evidence Schema

Validated Entities:
Ticker: $BIIB | Date: 2024-10-18 | Source: fda.gov/news-events/... | Event: Positive Panel Recommendation.

Formulated Hypothesis:

H0: Pre-announcement volume ratio (t-2 to t-1 vs 30d baseline) <= 1.05 at 95% confidence (α=0.05).

STAGE 4

Python Backtest Execution

Deterministic Validation

Sandboxed Code:
Queries Yahoo Finance historical OHLCV data. Calculates Abnormal Volume Ratio & Cumulative Abnormal Return (CAR).

Interactive Output:

Emits t-stat: 3.42 (p=0.002) with interactive chart, confidence band, and 1-click verifiable source audit trail.

Why TinyFish Powers the Core Engine

QuantIntel is purpose-built to leverage TinyFish's differentiated web agent capabilities.

TINYFISH SEARCH API

Low-Latency Live Web Discovery

- **Rank-Stable Indexing:** Retrieves authoritative, live financial announcements and earnings press releases without static knowledge cutoff limits.
- **Focused Domain Scoping:** Directs queries to regulatory portals (SEC EDGAR, FDA, FTC) and corporate investor relations desks.
- **Zero Token Waste:** Returns structured, ranked search hits rather than raw unstructured web sludge.

TINYFISH FETCH API

Pristine DOM-to-Markdown Extraction

- **Cleans Messy Web HTML:** Automatically strips navigation bars, ads, cookie popups, and boilerplate clutter.
- **80%+ Context Window Savings:** Delivers high-signal markdown directly into the LLM context, preventing token truncation and reducing latency.
- **Preserves Semantic Tables:** Maintains financial balance sheets, filing dates, and numerical disclosures with high fidelity.

TINYFISH AGENT RUNTIME

Autonomous Multi-Step Browser Execution

- **Dynamic JavaScript Execution:** Operates serverless Chromium to interact with modern SPAs that block traditional scrapers.
- **Interactive Navigation:** Autonomously clicks pagination controls, selects date ranges, and navigates deep corporate filing archives.
- **Stealth & Resiliency:** Built-in anti-bot protection ensures reliable extraction from institutional investor portals.

WHY TINYFISH IS CRITICAL: Standard search APIs (Tavily/Google) only return snippets and cannot interact with dynamic JS portals or bypass modern anti-bot systems. TinyFish acts as our full-fledged remote browser execution runtime, making live corporate deep research possible.

Multi-Tier Agentic Pipeline

A decoupled 3-tier architecture ensuring deterministic execution, strict security, and zero hallucination.

TIER 1 • INGESTION & WEB AGENTS

Live Discovery Layer

- **Streamlit Web Dashboard:** Interactive UI for analyst research prompt submission and live visual feedback.
- **Agent Orchestrator:** Decomposes research requests into targeted search terms and browsing strategies.
- **TinyFish Search & Fetch APIs:** Discovers live URLs, fetches raw HTML, and extracts clean markdown.
- **TinyFish Multi-Step Agent:** Executes headless Chromium sessions for deep JS interaction and form submissions.

TIER 2 • REASONING & SCHEMA CORE

Pydantic Verification Layer

- **LLM Reasoning Core:** High-precision entity extraction and evidence synthesis powered by GPT-4o / Claude 3.5.
- **Pydantic Contract Enforcement:** Forces structured outputs: verified event date, ticker, excerpt quote, and audited source URL.
- **Hypothesis Formulation Engine:** Translates qualitative event context into mathematically falsifiable null (H0) and alternative (H1) hypotheses.
- **Audit Trail Graph:** Maintains provenance linking every data point directly to its original TinyFish fetch URL.

TIER 3 • DETERMINISTIC EXECUTION

Python Quant Sandbox

- **Sandboxed Python Runtime:** Isolated execution environment (no external network or arbitrary code execution risks).
- **Historical Market Data:** Queries yfinance / Alpaca APIs for split-adjusted historical OHLCV pricing.
- **Event-Study Model Engine:** Computes Cumulative Abnormal Returns (CAR), volume surge z-scores, and t-test p-values.
- **Visual Research Brief:** Outputs interactive Matplotlib/Plotly figures and downloadable markdown summaries.

SECURITY & DATA FIDELITY: Python execution is strictly sandboxed with read-only market data connectors. No live broker API keys or trade execution capabilities are included, ensuring zero financial execution hazard or unauthorized code execution.

Competitive Matrix & Defensibility

QuantIntel occupies a unique white space between general AI chatbots and rigid quant platforms.

Key Capabilities	General LLMs (ChatGPT / Claude)	Search Engines (Perplexity / Google)	Quant Platforms (QuantConnect / LEAN)	QuantIntel (Our Solution)
Live Web Navigation	❌ Static Weights (Cutoff)	⚠️ Basic Web Search Only	❌ None (Data must be uploaded)	✅ **Native TinyFish Agent**
Dynamic JS & Anti-Bot	❌ Cannot interact	❌ Fails on complex SPAs	❌ N/A	✅ **Full Chromium Browser Fleet**
Source Provenance	❌ Unaudited / Hallucinated	⚠️ Text Citations Only	❌ None	✅ **Exact Quotes & Sourced URLs**
Mathematical Precision	❌ Hallucinates Arithmetic	❌ Hallucinates Math	✅ Deterministic Code	✅ **Sandboxed Python Engine**
Hypothesis Generation	⚠️ Qualitative Speculation	⚠️ Search Summaries	❌ Manual Code Writing Only	✅ **Automated Falsifiable H0/H1**
Target Cost / Accessibility	Free / $20/mo	$20/mo	Developer-Heavy	⚡ **Accessible, API-Driven SaaS**

THE QUANTINTEL MOAT: We bridge the chasm between messy live-web reconnaissance and deterministic quantitative finance. General LLMs fail on numbers; quant platforms fail on live text. QuantIntel marries TinyFish web discovery with rigorous Python backtesting.

Hackathon MVP Scope vs. Future Horizon

A laser-focused, 100% executable prototype built for a rock-solid hackathon demonstration.

48-HOUR HACKATHON MVP

What the Judges Will See in the Live Demo

✅ **Streamlit Web Dashboard:** Intuitive search bar allowing analysts to enter natural-language research queries with live status logs.

✅ **Live TinyFish Agent Tracing:** Real-time visual display showing TinyFish Search & Fetch discovering URLs and cleaning DOM data.

✅ **Pydantic Evidence Card:** Structured summary extracting event dates, verifiable claims, and direct clickable source URLs.

✅ **Automated Hypothesis Engine:** Translates raw event into a measurable statistical test (e.g., Abnormal Volume Surge around earnings).

✅ **Sandboxed Python Event Study:** Queries Yahoo Finance historical data and runs an empirical event study backtest.

✅ **Interactive CAR Visuals:** Renders Cumulative Abnormal Returns, confidence bands, t-statistics, and downloadable PDF research brief.

DELIBERATELY OUT OF SCOPE

Disciplined Guardrails for Hackathon Success

🚫 **Live Order Execution / Auto-Trading:** Zero capital allocation or broker connectivity. Eliminates regulatory and financial catastrophe risks.

🚫 **Microsecond / HFT Low-Latency:** QuantIntel targets discretionary, fundamental, and event-driven quants, not sub-millisecond market-making.

🚫 **Guaranteed Price Predictions:** We reject snake-oil financial promises. The agent proposes falsifiable hypotheses; it does not promise profits.

🚫 **Tick-Level Order Book Ingestion:** Scope is constrained to daily and hourly OHLCV bars suitable for event studies.

🚫 **Full Autonomous Portfolio Management:** Human-in-the-loop design: The system acts as a research accelerator for human decision-makers.

OUR PRODUCT PHILOSOPHY: “A focused, bulletproof 5-minute research loop that reliably works live on stage is worth 100x more than 20 half-finished speculative features.” Every feature built is designed for reproducible live demo verification.

Evaluation Framework & Responsible AI

Institutional-grade rigor and ethical safeguards embedded into every layer of QuantIntel.

CRITERION 1

Source Grounding & Provenance

100% Sourced Citations

✓ **Verifiable Evidence:** Every factual statement and event date must trace directly to an audited TinyFish search URL.

✓ **Verbatim Context Matching:** The system highlights the exact extracted sentence from the source markdown.

✓ **Zero Synthetic Data:** If an event date or ticker is ambiguous, the agent flags it as missing rather than hallucinating.

CRITERION 2

Temporal Integrity & Anti-Leakage

0% Look-Ahead Bias

✓ **UTC Timestamp Validation:** Event announcement timestamps are strictly verified against official press timestamps.

✓ **Pre/Post Window Isolation:** Statistical backtesting windows are isolated to ensure no future data leaks into the estimation window.

✓ **Survivorship Awareness:** Flags corporate delisting or restructuring caveats alongside backtest metrics.

CRITERION 3

Statistical Reproducibility

Open-Source Math

✓ **Transparent Python Code:** Every backtest generates clean, comment-annotated Python code downloadable with the brief.

✓ **Standard Statistical Metrics:** Reports standard Student's t-statistics, p-values, and 95% confidence intervals.

✓ **Sensitivity Analysis:** Tests robustness across varying estimation windows (-30d to -120d) to prevent p-hacking.

REGULATORY & ETHICAL SAFEGUARD: QuantIntel is strictly an analytical research assistant and explicitly disclaims financial advisory status. All generated reports include clear methodological caveats, uncertainty bounds, and explicit risk disclosures in alignment with institutional compliance standards.

Sprint Roadmap to TinyFish Accelerator

From hackathon prototype to scalable commercial SaaS platform in the TinyFish ecosystem.

PHASE 1 • HOURS 0-12

TinyFish Recon Pipeline

- Integrate TinyFish Search & Fetch APIs
- Test dynamic JS corporate portal crawling
- Implement Pydantic schema validation
- Verify source URL and timestamp parsing

PHASE 2 • HOURS 12-24

Quant Backtest Sandbox

- Build sandboxed Python event-study runner
- Connect yfinance historical pricing API
- Implement CAR and volume z-score math
- Validate hypothesis generation templates

PHASE 3 • HOURS 24-36

Interactive UI & Visuals

- Develop Streamlit analyst web dashboard
- Display live TinyFish browser trace logs
- Render interactive Plotly CAR chart
- Add downloadable research brief exporter

PHASE 4 • HOURS 36-48

Demo Polish & Pitch

- End-to-end edge case stress testing
- Test regulatory dockets (SEC & FDA)
- Record 3-minute video pitch & demo
- Prepare submission for TinyFish Accelerator

POST-HACKATHON ACCELERATOR TRAJECTORY: Our goal is the TinyFish $2M Pre-Accelerator Golden Ticket. Post-hackathon, we will commercialize QuantIntel as a B2B research copilot for boutique hedge funds, family offices, and equity analysts—monetized via tiered monthly subscriptions and TinyFish agent execution credits.